

EMPLOYEE ATTRITION AND PERFORMANCE ANALYSIS

Internship Project Report

Submitted By

SALONI BANSOD

Submitted To

SLASH MARK IT SOLUTIONS

Tools Used:

Microsoft Excel

SQL

Python

Machine Learning

Dataset:

IBM HR Analytics Employee Attrition Dataset

Year: 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my internship organization for providing me with the opportunity to work on this project. This internship helped me gain practical knowledge of data analysis and machine learning.

I would also like to thank my mentor and faculty members for their continuous guidance, encouragement, and support throughout the project.

The experience gained while working on this project has enhanced my technical skills in Excel, SQL, Python, and Machine Learning and has helped me understand the practical applications of data analytics.

I am grateful to everyone who contributed directly or indirectly to the successful completion of this project.

ABSTRACT

Employee attrition is a major concern for organizations as it affects productivity, employee morale, and operational costs. This project focuses on analyzing employee attrition using the IBM HR Analytics Employee Attrition dataset.

The analysis was performed using Microsoft Excel, SQL, Python, and Machine Learning techniques. Excel was used for data cleaning and visualization, SQL was used for querying and retrieving insights, and Python was used for exploratory data analysis and predictive modeling.

A Random Forest Classifier was implemented to predict employee attrition based on various employee-related factors. The project identified important factors influencing attrition such as overtime, department, salary, and employee demographics.

The findings of this project can help organizations make informed decisions and improve employee retention strategies.

INTRODUCTION

Employee attrition refers to the reduction in the workforce of an organization due to employees leaving their jobs. Employees may leave an organization because of retirement, resignation, personal reasons, better career opportunities, dissatisfaction, or work-related issues. High employee attrition can negatively affect organizational productivity and increase recruitment and training costs.

In today's competitive business environment, retaining talented employees has become one of the most important challenges for organizations. Understanding the reasons behind employee attrition can help companies develop effective strategies to improve employee satisfaction and retention.

This project focuses on analyzing employee attrition using the IBM HR Analytics Employee Attrition dataset. The project uses data analysis techniques to identify patterns and factors that influence employee

turnover. Microsoft Excel, SQL, Python, and Machine Learning techniques were used to analyze the dataset and generate meaningful insights.

The project also includes the development of a predictive model using the Random Forest algorithm to predict employee attrition. The results obtained from this analysis can help organizations make informed decisions and improve workforce management practices.

OBJECTIVES

The main objectives of this project are:

1. To analyze employee attrition using historical HR data.
2. To identify factors that influence employee turnover.
3. To perform data cleaning and preprocessing on the dataset.
4. To create visualizations and reports using Microsoft Excel.
5. To retrieve meaningful insights using SQL queries.
6. To perform exploratory data analysis using Python.
7. To build a Machine Learning model for predicting employee attrition.
8. To provide recommendations that can help organizations improve employee retention.

DATASET DESCRIPTION

The dataset used in this project is the IBM HR Analytics Employee Attrition Dataset. This dataset contains employee-related information that can be used to analyze employee behavior and attrition patterns.

The dataset consists of 1470 employee records and 35 attributes. Each record represents an individual employee and includes details such as age, department, education, job role, monthly income, overtime status, years of experience, and attrition status.

Some important attributes included in the dataset are:

- Age
- Attrition
- Department
- Education
- Gender
- Job Role
- Monthly Income
- OverTime

- Years at Company
- Total Working Years

The dataset provides valuable information that helps in understanding the factors influencing employee attrition and employee performance.

df.head()

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	

5 rows x 35 columns

df.shape

(1470, 35)

DATA CLEANING

Data cleaning is an important step in any data analysis project because it ensures the accuracy and reliability of the results.

The dataset was examined for missing values, duplicate records, and inconsistencies before performing analysis. Missing value analysis was conducted using both Excel and Python. The results showed that there were no significant missing values present in the dataset.

Duplicate records were also checked to ensure that each employee record was unique. No duplicate records affecting the analysis were identified.

After confirming the quality of the dataset, the data was prepared for further analysis and visualization using Excel, SQL, and Python.

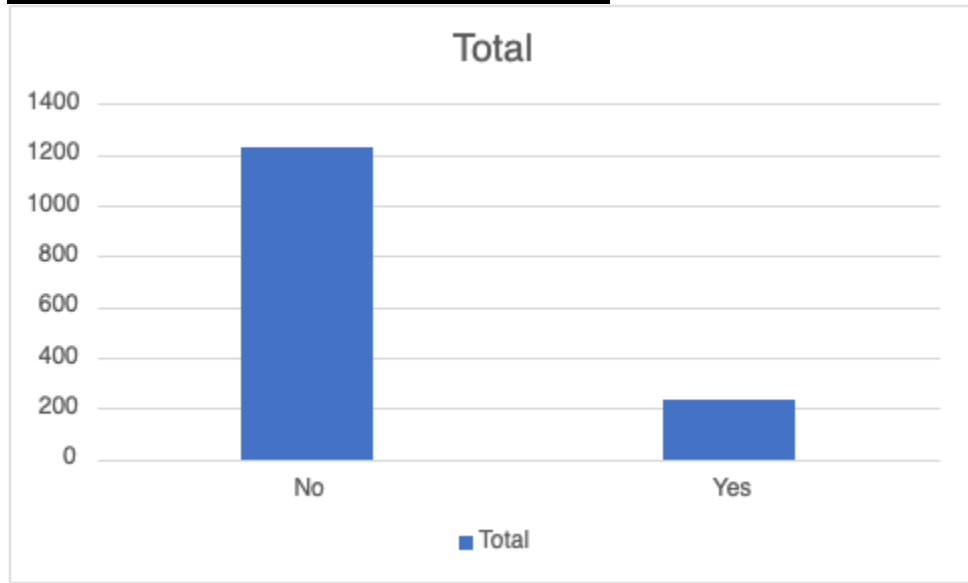
```
[6] df.info()
✓ Os
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                    1470 non-null   int64
1   Attrition                            1470 non-null   object
2   BusinessTravel                       1470 non-null   object
3   DailyRate                            1470 non-null   int64
4   Department                           1470 non-null   object
5   DistanceFromHome                    1470 non-null   int64
6   Education                            1470 non-null   int64
7   EducationField                       1470 non-null   object
8   EmployeeCount                       1470 non-null   int64
9   EmployeeNumber                      1470 non-null   int64
10  EnvironmentSatisfaction              1470 non-null   int64
11  Gender                               1470 non-null   object
12  HourlyRate                           1470 non-null   int64
13  JobInvolvement                      1470 non-null   int64
14  JobLevel                             1470 non-null   int64
15  JobRole                              1470 non-null   object
16  JobSatisfaction                     1470 non-null   int64
17  MaritalStatus                       1470 non-null   object
18  MonthlyIncome                       1470 non-null   int64
19  MonthlyRate                          1470 non-null   int64
20  NumCompaniesWorked                  1470 non-null   int64
```

Employee Attrition Analysis

A Pivot Table was created in Microsoft Excel to analyze employee attrition. The analysis showed that the majority of employees remained with the company, while a smaller percentage left the organization.

This indicates that employee retention was generally high; however, understanding the reasons behind employee attrition remains important for organizational growth and workforce planning.

Attrition	Count of EmployeeNumber
No	1233
Yes	237
Grand Total	1470



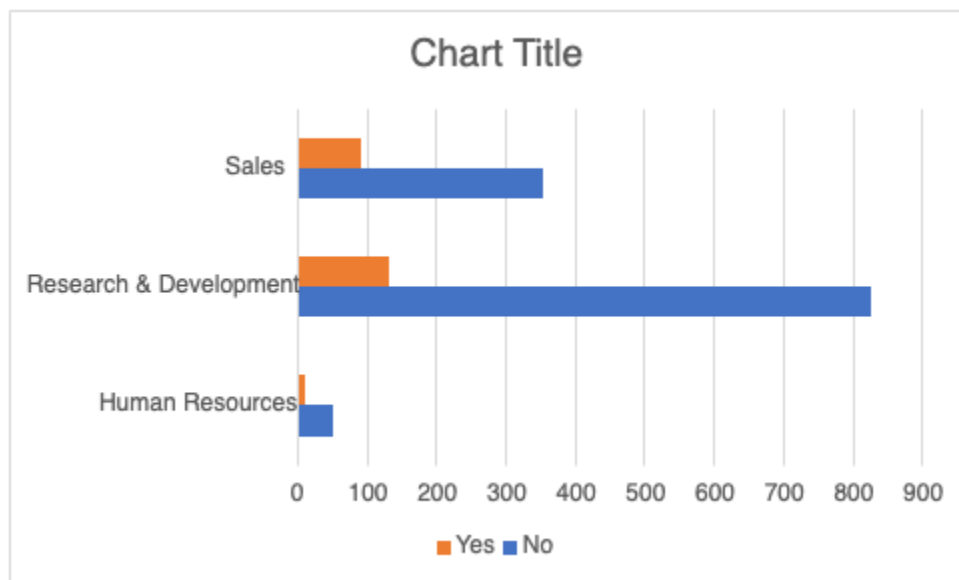
10.1 Department-wise Attrition Analysis

Department-wise analysis was performed using Pivot Tables and charts in Microsoft Excel. The objective was to identify which departments experienced higher employee attrition.

The analysis revealed that employees from the Research & Development and Sales departments showed relatively higher attrition compared to other departments. This may be due to workload, job pressure, career growth opportunities, or other organizational factors.

Understanding department-wise attrition helps organizations focus their retention strategies on specific departments where employee turnover is more frequent.

Count of EmployeeNumber	Attrition		
Department	No	Yes	Grand Total
Human Resources	51	12	63
Research & Development	828	133	961
Sales	354	92	446
Grand Total	1233	237	1470



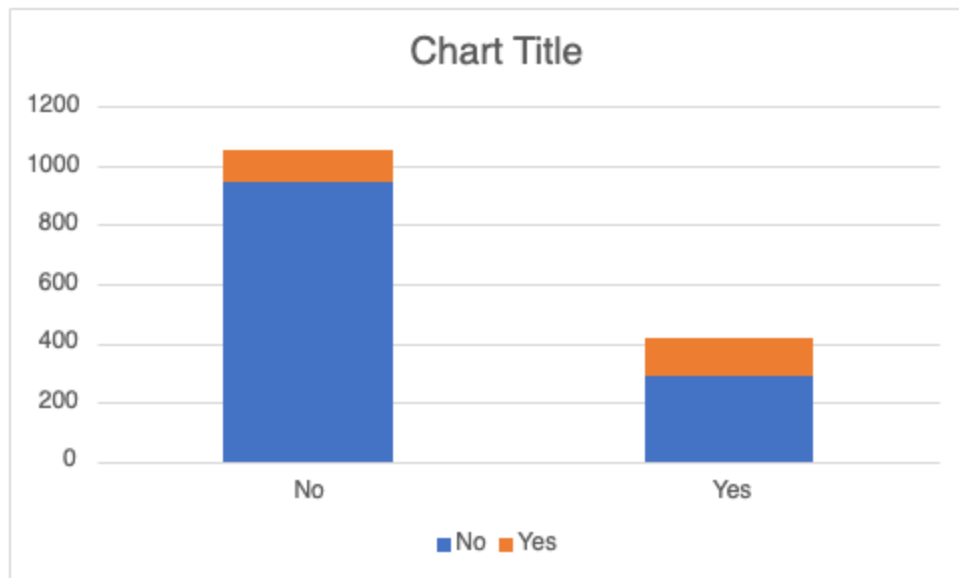
10.2 Overtime vs Attrition Analysis

Overtime is considered one of the important factors affecting employee satisfaction and retention. To analyze its impact, a Pivot Table and chart were created comparing overtime status with attrition.

The analysis showed that employees who frequently worked overtime were more likely to leave the organization than employees who did not work overtime.

This finding suggests that maintaining a healthy work-life balance can play an important role in reducing employee turnover and improving employee satisfaction.

Count of EmployeeNumber	Attrition		
OverTime	No	Yes	Grand Total
No	944	110	1054
Yes	289	127	416
Grand Total	1233	237	1470



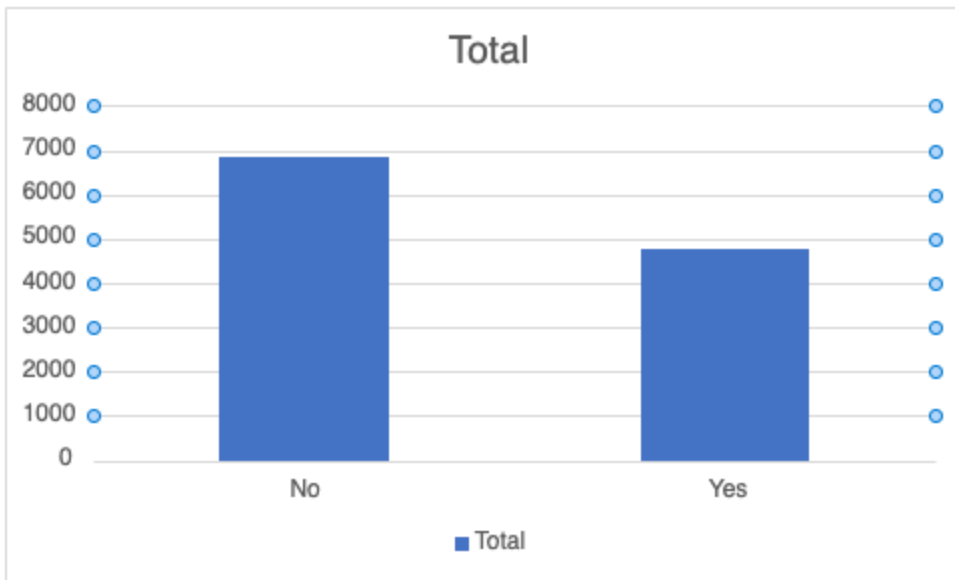
10.3 Salary Analysis

Salary is one of the major factors influencing employee retention. A Pivot Table was created to compare the average monthly income of employees who stayed with the organization and those who left.

The analysis indicated that salary levels have a noticeable impact on employee attrition. Employees with lower salaries showed a higher tendency to leave the organization in search of better opportunities.

Organizations can improve employee retention by offering competitive compensation packages and performance-based rewards.

Attrition	Average of MonthlyIncome
No	6832.739659
Yes	4787.092827
Grand Total	6502.931293



SQL ANALYSIS

Structured Query Language (SQL) was used to retrieve important insights from the employee attrition dataset. SQL queries help in extracting, filtering, and summarizing data efficiently.

The following queries were used during the analysis.

```
SELECT COUNT(*) AS TotalEmployees
```

```
FROM employee_attrition;
```

```
SELECT Attrition, COUNT(*) AS EmployeeCount
```

```
FROM employee_attrition
```

```
GROUP BY Attrition;
```

```
SELECT Department, COUNT(*) AS AttritionCount
```

```
FROM employee_attrition
```

```
WHERE Attrition='Yes'
```

```
GROUP BY Department;
```

```
SELECT AVG(MonthlyIncome) AS AverageSalary
```

```
FROM employee_attrition
```

```
WHERE Attrition='Yes';
```

```
SELECT OverTime, COUNT(*) AS AttritionCount
```

```
FROM employee_attrition
```

```
WHERE Attrition='Yes'
```

```
GROUP BY OverTime;
```


PYTHON ANALYSIS

Python was used for Exploratory Data Analysis (EDA) and visualization. Libraries such as Pandas, Matplotlib, and Seaborn were used to analyze employee data and identify trends.

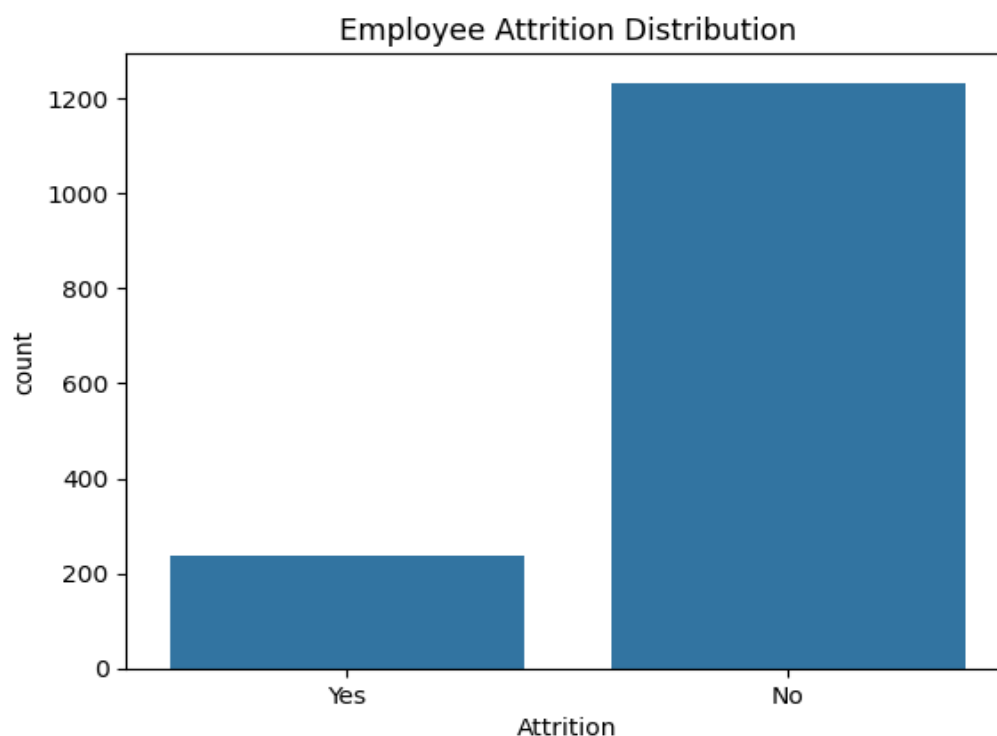
The dataset was loaded into Python using Pandas. Dataset information, missing values, and statistical summaries were examined before creating visualizations.

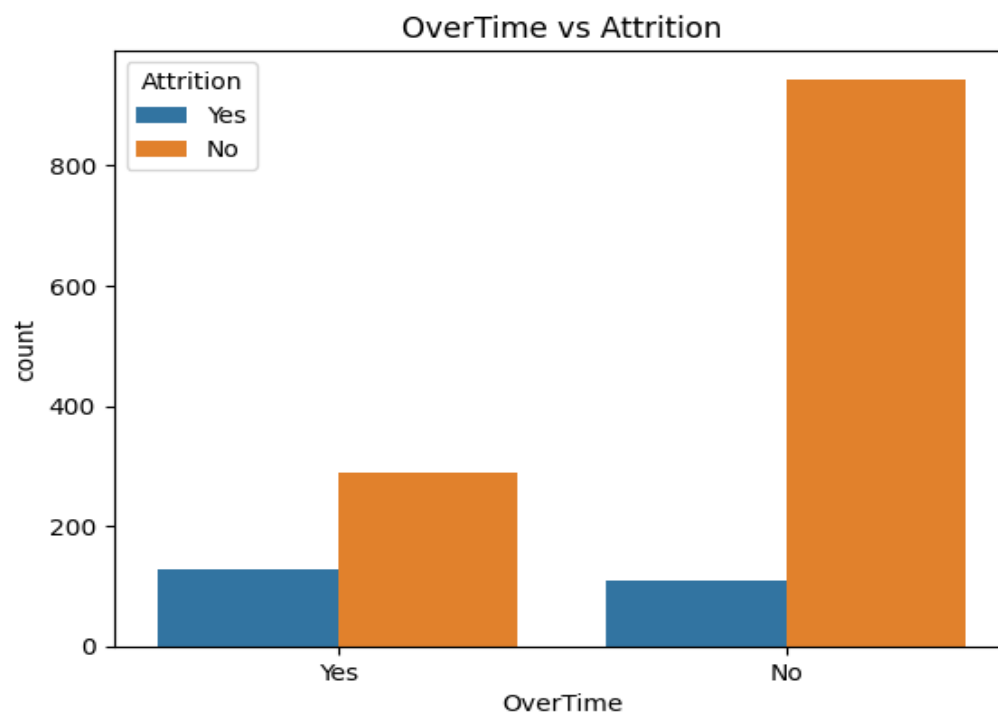
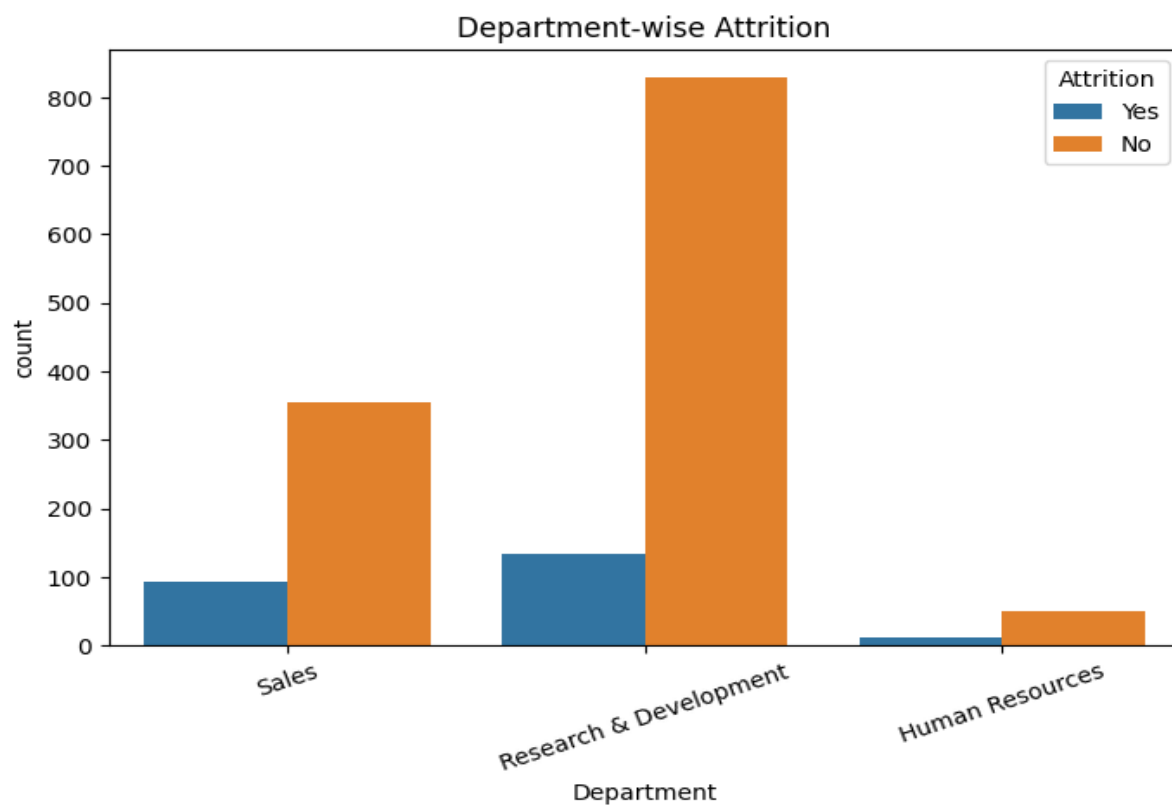
Several graphs were created to understand employee attrition patterns, department-wise attrition, overtime impact, and age distribution.

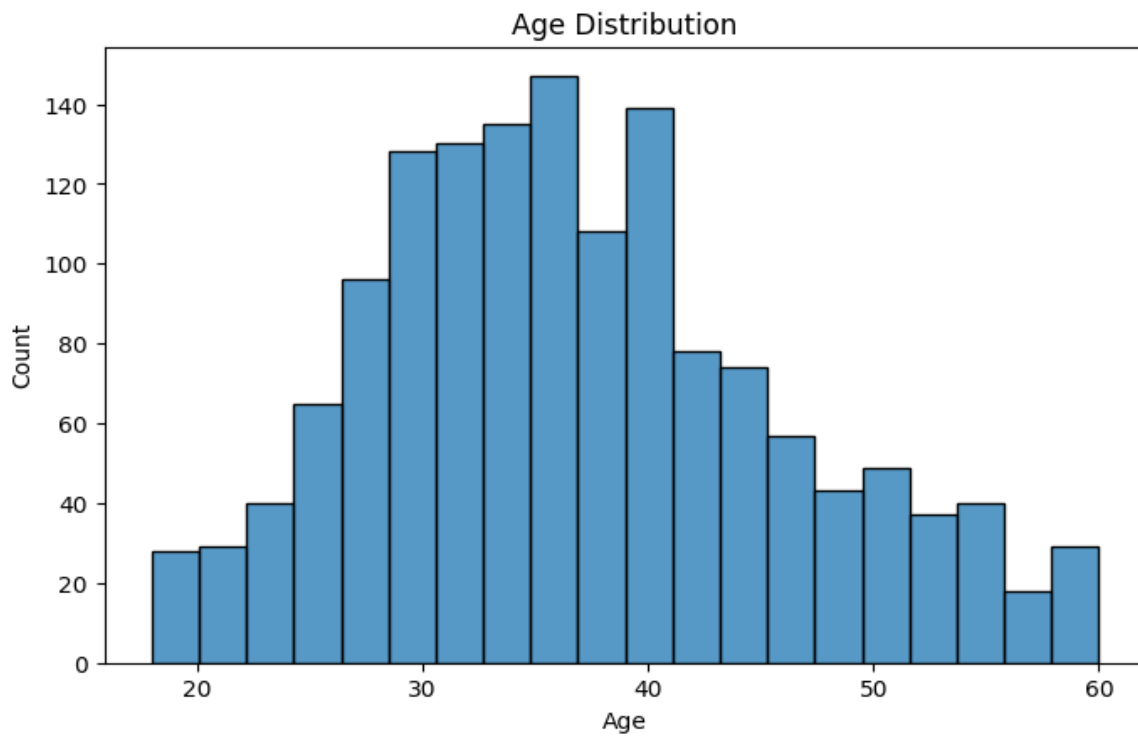
Data visualization plays an important role in understanding patterns and relationships within a dataset. Various graphs were created using Python to visually represent employee attrition trends.

The visualizations provided a better understanding of employee behavior and helped identify factors influencing employee turnover.

The graphs confirmed that overtime, salary, and department were significant factors affecting employee attrition.







MACHINE LEARNING MODEL

Machine Learning was used to predict employee attrition based on various employee-related factors. The Random Forest Classifier algorithm was selected because it provides good accuracy and handles complex datasets effectively.

Before training the model, categorical variables were converted into numerical form using Label Encoding and One-Hot Encoding techniques. The dataset was then divided into training and testing datasets using an 80:20 ratio.

The Random Forest model was trained using the training dataset and tested on the testing dataset. Predictions were generated and compared with actual values to evaluate model performance.

The model achieved satisfactory accuracy and successfully predicted employee attrition based on employee characteristics and workplace factors.

```

from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Attrition'] = le.fit_transform(df['Attrition'])

df['Attrition'].head()

```

```

...
Attrition
0      1
1      0
2      1
3      0
4      0

```

dtype: int64

```
df = pd.get_dummies(df, drop_first=True)
```

```
df.head()
```

```

...
Age  Attrition  DailyRate  DistanceFromHome  Education  EmployeeCount  EmployeeNumber  EnvironmentSatisfaction  HourlyRate  JobI
0    41         1      1102             1         2         1         1         2         94
1    49         0      279             8         1         1         2         3         61
2    37         1     1373             2         2         1         4         4         92
3    33         0     1392             3         4         1         5         4         56
4    27         0      591             2         1         1         7         1         40

```

5 rows x 48 columns

```

X = df.drop('Attrition', axis=1)
y = df['Attrition']

```

```

print(X.shape)
print(y.shape)

```

```

(1470, 47)
(1470,)

```

```
s from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
s print(X_train.shape)
print(X_test.shape)
```

```
(1176, 47)
(294, 47)
```

```
s  from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)
```

```
...  RandomForestClassifier  
RandomForestClassifier(random_state=42)
```

```
s pred = model.predict(X_test)

print(pred[:10])

[0 0 0 0 0 0 0 0 0 0]
```

```
s from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, pred)

print("Accuracy =", accuracy)

Accuracy = 0.8775510204081632
```

```
s  from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, pred)

print(cm)

... [[254  1]
     [ 35  4]]
```

FORECAST OF HR DYNAMICS (5–10 YEARS)

Based on the analysis of historical employee data, it is expected that employee attrition will continue to be influenced by factors such as overtime, salary levels, job satisfaction, and career growth opportunities.

If organizations fail to address these factors, employee turnover may increase over time, leading to higher recruitment and training costs. Departments experiencing high attrition may face workforce shortages and productivity challenges.

Over the next five to ten years, organizations are expected to increasingly use data analytics and machine learning techniques to monitor employee behavior and identify employees who may be at risk of leaving.

By improving employee engagement, maintaining work-life balance, providing competitive compensation, and creating clear career growth opportunities, organizations can significantly reduce future attrition and improve employee retention.

KEY FINDINGS

The following findings were obtained from the analysis:

1. The dataset contained 1470 employee records and 35 attributes.
2. Most employees remained with the company, while a smaller number left the organization.
3. Employees working overtime showed higher attrition compared to employees who did not work overtime.
4. Research & Development and Sales departments experienced relatively higher employee turnover.
5. Salary levels had a significant impact on employee retention.
6. Employee attrition was influenced by multiple factors including overtime, department, income, and work experience.
7. The Random Forest Machine Learning model successfully predicted employee attrition with good accuracy.
8. Data analytics techniques helped identify important patterns that can support better workforce management decisions.

RECOMMENDATIONS

Based on the findings of this project, the following recommendations are suggested:

1. Reduce excessive overtime and encourage a healthy work-life balance.
2. Offer competitive salary packages and performance-based incentives.
3. Provide career development and training opportunities for employees.
4. Conduct regular employee satisfaction surveys to identify workplace concerns.
5. Improve employee engagement through recognition programs and team-building activities.
6. Focus on departments with higher attrition and implement targeted retention strategies.
7. Utilize predictive analytics and machine learning models to identify employees who may be at risk of leaving.
8. Continuously monitor workforce trends to improve employee retention and organizational performance.

CONCLUSION

This project successfully analyzed employee attrition using the IBM HR Analytics Employee Attrition dataset. Microsoft Excel, SQL, Python, and Machine Learning techniques were used to examine employee behavior and identify factors contributing to employee turnover.

The analysis revealed that overtime, salary, and department were among the most significant factors affecting employee attrition. Various visualizations and statistical analyses were performed to better understand employee retention patterns.

A Random Forest Classifier was developed to predict employee attrition and achieved satisfactory accuracy. The results demonstrated the effectiveness of data analytics and machine learning in solving real-world human resource challenges.

Overall, this project provided valuable insights into employee attrition and highlighted the importance of data-driven decision-making for improving workforce management and employee retention.